

A Multi-Benchmark Stochastic Dominance Approach to Enhanced Indexation

Antonio Esteves^a, Cristiano Arbex^a, Francesco Cesarone^b

^a*Universidade Federal de Minas Gerais
Departamento de Ciencia da Computação*

^b*Roma Tre University
Department of Business Studies*

Abstract

Keywords:

1. Preliminary Concepts

Stochastic dominance criteria are particularly attractive for portfolio selection because they establish a partial ordering among random variables without requiring the specification of a particular utility function to represent an investor's preferences (Whitmore and Findlay, 1978). The concept of stochastic dominance was first introduced by Blackwell (1953) and, although it appeared in early work in statistics and economic theory, it was only more fully developed in the late 1960s and early 1970s (Hadar and Russell, 1969; Hanoch and Levy, 1969; Rothschild and Stiglitz, 1970; Whitmore, 1970). Since then, the literature has expanded considerably, with hundreds of studies examining the topic, among which the survey by Kroll and Levy (1980) and the bibliography compiled by Bawa (1982) remain influential references.

The Expected Utility Theory (EUT) was formally introduced in the seminal work of von Neumann and Morgenstern (1947) and has since been extensively developed by numerous scholars. However, the conceptual foundations of utility theory trace back to the 18th century, particularly to the contributions of the mathematician Bernoulli (1954). Despite its limitations, expected utility theory remains the predominant framework in economic analysis for modeling the preferences and decision-making behavior of rational agents. Its widespread adoption stems from its axiomatic foundation and its ability to provide a systematic approach to decision-making under uncertainty.

The expected utility framework is based on the assumption that investors act rationally, meaning they generally prefer greater wealth over less. However, psychologists have challenged this notion, highlighting experimental evidence that investors do not always behave in a fully rational manner. Moreover, in certain situations, investors systematically alter objective probabilities, raising questions about the reliability of the expected utility model. These experimental insights are valuable as they help clarify investor behavior anomalies and explain deviations from the predicted equilibrium prices of risky assets under the expected utility framework.

The notion of evaluating random variables based on their expected utility was initially proposed by Bernoulli (1954) and over the last century, expected utility theory has become

particularly influential in economic contexts, where rational preference axioms serve as its foundation. The concept of expected utility was further formalized in decision theory by von Neumann and Morgenstern (1947) and provides the basis for extending a utility function defined on real numbers (outcomes) to one that applies to random variables. Each random variable is assigned a utility value that reflects both the utility of its possible outcomes and the probabilities associated with those outcomes.

Let X and Y be two random variables with cumulative distribution functions $F_X(\alpha) = \Pr(X \leq \alpha)$ and $F_Y(\alpha) = \Pr(Y \leq \alpha)$ for $\alpha \in \mathbb{R}$.

Definition 1 (First-order Stochastic Dominance — FSD). X stochastically dominates Y in first order ($X \succeq_{FSD} Y$) iff:

$$F_X(\alpha) \leq F_Y(\alpha), \quad \forall \alpha \in \mathbb{R}. \quad (1)$$

Definition 2 (Second-order Stochastic Dominance — SSD). X stochastically dominates Y in second order ($X \succeq_{SSD} Y$) iff:

$$\int_{-\infty}^{\alpha} F_X(\tau) d\tau \leq \int_{-\infty}^{\alpha} F_Y(\tau) d\tau, \quad \forall \alpha \in \mathbb{R}. \quad (2)$$

Both two definitions require that each inequality is strict in at least one case.

As discussed in Levy (1992), these relations can be interpreted within the framework of EUT through different classes of utility functions. In particular, X is preferred to Y with respect to FSD if and only if $E[u(X)] \geq E[u(Y)]$ for all non-decreasing utility functions u , whereas X is preferred to Y with respect to SSD if and only if the same condition holds for all non-decreasing and concave utility functions. Compared with FSD, the SSD relation is less restrictive. More generally, for a stochastic dominance criterion of order v , the condition for $(v+1)$ -SD becomes progressively less restrictive as the order increases. Moreover, v -SD implies $(v+1)$ -SD, while the converse does not necessarily hold (Levy, 2006).

We denote by R_I the return of the financial index, represented by S equally likely outcomes, whose ordered values are given by $R_I^{(1)} \leq \dots \leq R_I^{(S)}$. Similarly, let R denote a random return with ordered outcomes $R^{(1)} \leq \dots \leq R^{(S)}$. Assume that, for discrete distributions, the scenarios for both the portfolio return R and the index return R_I occur with equiprobable scenarios, the first- and second-order stochastic dominance relations can then be expressed as follows:

(a) R dominates R_I with respect to FSD ($R \succeq_{FSD} R_I$) if and only if:

$$R^{(i)} \geq R_I^{(i)}, \quad i = 1, \dots, S.$$

(b) R dominates R_I with respect to SSD ($R \succeq_{SSD} R_I$) if and only if:

$$\sum_{j=1}^i R^{(j)} \geq \sum_{j=1}^i R_I^{(j)}, \quad i = 1, \dots, S.$$

The CVaR of a random return R at confidence level $\alpha \in (0, 1)$ provides a mathematical formalization of the notion of the “average loss” over the worst $\alpha \times 100\%$ outcomes (Acerbi and Tasche, 2002), where losses are measured relative to zero payoffs. Note that is straightforward to see that, under equiprobable scenarios,

$$CVaR_\alpha(R) = -\frac{1}{\alpha} \sum_{j=1}^i R^{(j)}, \forall i \in \{1 \dots S\}$$

, where $CVaR_\alpha$ represents the Conditional Value-at-Risk at a confidence level of α . Then, we have that

$$Tail_\alpha(R) = -\alpha CVaR_\alpha(R).$$

Thus, requiring $Tail_\alpha(R) \geq Tail_\alpha(R_I)$ is equivalent to imposing

$$CVaR_\alpha(R) \leq CVaR_\alpha(R_I)$$

for all $\alpha \in (0, 1)$

In the particularly interesting case, of equiprobable scenarios where $1/S$ for all i , the above criteria simplify, resulting in:

(a) R dominates R_I with respect to FSD ($R \succeq_{\text{FSD}} R_I$) if and only if:

$$R^{(i)} \geq R_I^{(i)}, \quad i = 1, \dots, S.$$

(b) R dominates R_I with respect to SSD ($R \succeq_{\text{SSD}} R_I$) if and only if:

$$\sum_{j=1}^i R^{(j)} \geq \sum_{j=1}^i R_I^{(j)}, \quad i = 1, \dots, S.$$

Also in this case, $Tail_{\frac{1}{S}}(R) = \frac{1}{S} \sum_{j=1}^i R^{(j)}$. Next, since

$$CVaR_{\frac{1}{S}}(R) = -\frac{1}{j} \sum_{j=1}^i R^{(j)}$$

we have that $CVaR_{\frac{1}{S}}(R) = -\frac{1}{j} \sum_{j=1}^i R^{(j)}$. Thus, requiring $Tail_{\frac{1}{S}}(R) \geq Tail_{\frac{1}{S}}(R_I)$ is equivalent to considering $CVaR_{\frac{1}{S}}(R) \leq CVaR_{\frac{1}{S}}(R_I) \quad \forall i \in \{1 \dots S\}$.

1.1. Multi-objective framework

Following Roman et al. (2006), the portfolio selection problem can be cast naturally as a multi-objective optimisation problem. Given S equally probable scenarios and a feasible set X of portfolio weights, one seeks a portfolio $x \in X$ whose return R_x is as large as possible under every scenario simultaneously. Denoting $f_i(x) = \sum_{j=1}^N r_{ij}w_j$ as the portfolio return under scenario $i \in \{1, \dots, S\}$, this reads:

$$\max \{f(x) = (f_1(x), \dots, f_S(x)) : x \in X\}.$$

An achievement vector $y = (y_1, \dots, y_S) \in \mathbb{R}^S$ *Pareto dominates* $y' = (y'_1, \dots, y'_S)$ if $y_i \geq y'_i$ for all i , with at least one strict inequality. A feasible portfolio x_0 is *Pareto efficient* if no other $x \in X$ produces a Pareto dominant achievement vector.

The connection to SSD efficiency arises from the fact that comparing two discrete distributions under SSD is equivalent to comparing the cumulative sums of their ordered outcomes. Roman et al. (2006) establish that, for the sum of the k worst portfolio returns under scenario ordering, this quantity equals the optimal value of the linear programme:

$$\max \left(kt_k - \sum_{i=1}^S d_{ki} \right) \quad \text{subject to} \quad t_k - f_i(x) \leq d_{ki}, \quad d_{ki} \geq 0, \quad \forall i = 1, \dots, S.$$

Using this representation for every $k = 1, \dots, S$ yields the following multi-objective linear programme, whose Pareto efficient solutions are exactly the SSD efficient portfolios:

$$\max \left(t_1 - \sum_{i=1}^S d_{1i}, \quad 2t_2 - \sum_{i=1}^S d_{2i}, \quad \dots, \quad St_S - \sum_{i=1}^S d_{Si} \right) \quad (3)$$

subject to:

$$t_k - f_i(x) \leq d_{ki}, \quad d_{ki} \geq 0 \quad \forall i, k = 1, \dots, S, \quad \sum_{j=1}^N w_j = 1, \quad w_j \geq 0.$$

Proposition (Roman et al., 2006). *A portfolio x^* is SSD efficient if and only if there exist vectors t^* and $d^* \geq 0$ such that (x^*, t^*, d^*) is a Pareto efficient solution of (3).*

An equivalent and more compact reformulation follows directly from the CVaR representation of SSD. As shown in Roman et al. (2013), the SSD condition $\text{CVaR}_{i/S}(R^x) \leq \text{CVaR}_{i/S}(R_I)$ holding for all $i = 1, \dots, S$ can be expressed in terms of tail expectations, so that SSD efficient portfolios are the Pareto efficient solutions of:

$$\max \left(\text{Tail}_{\frac{1}{S}}(R^x), \text{Tail}_{\frac{2}{S}}(R^x), \dots, \text{Tail}_{\frac{S}{S}}(R^x) \right), \quad x \in X. \quad (4)$$

The i -th objective $\text{Tail}_{i/S}(R^x) = \frac{1}{S} \sum_{j=1}^i R^{x(j)}$ is the mean of the i worst portfolio returns, capturing the lower tail of the return distribution at resolution i/S . Improving any one tail without worsening another corresponds precisely to an improvement in SSD, so the Pareto front of (4) sweeps out the entire SSD efficient frontier.

1.2. Roman scalarization approach

Computing the full Pareto front of (4) is generally intractable. Roman et al. (2006) reduce it to a single-objective problem via the *reference point method* of ?, an interactive technique for multi-objective programming in which a user-specified aspiration vector guides the search toward a preferred Pareto efficient solution.

The aspiration vector is set to the tail profile of the benchmark index R_I :

$$\hat{\tau} = (\hat{\tau}_1, \dots, \hat{\tau}_S) := \left(\text{Tail}_{\frac{1}{S}}(R_I), \dots, \text{Tail}_{\frac{S}{S}}(R_I) \right), \quad \hat{\tau}_i = \frac{1}{S} \sum_{j=1}^i R_I^{(j)}.$$

Each component $\hat{\tau}_i$ is the mean of the i worst index returns; the vector $\hat{\tau}$ thus encodes the full distributional shape of the benchmark from the left tail to the mean. The *achievement function* measures the worst-case shortfall of the portfolio tails relative to these aspiration levels:

$$\Gamma_{\hat{\tau}} \left(\text{Tail}_{\frac{1}{S}}(R^x), \dots, \text{Tail}_{\frac{S}{S}}(R^x) \right) = \min_{1 \leq i \leq S} \left(\text{Tail}_{\frac{i}{S}}(R^x) - \hat{\tau}_i \right).$$

Maximising $\Gamma_{\hat{\tau}}$ over $x \in X$, and introducing the scalar $\vartheta = \min_i (\text{Tail}_{i/S}(R^x) - \hat{\tau}_i)$, yields the *unscaled model* (Roman et al., 2006, 2013):

$$\max \vartheta \quad \text{subject to} \quad \vartheta \leq \text{Tail}_{\frac{i}{S}}(R^x) - \hat{\tau}_i, \quad i = 1, \dots, S, \quad x \in X. \quad (5)$$

A small augmentation term $\varepsilon \sum_{i=1}^S (\text{Tail}_{i/S}(R^x) - \hat{\tau}_i)$ is usually added to the objective to guarantee strict Pareto efficiency of the optimum (Roman et al., 2006). The model (5) is always feasible and its optimal solution is SSD efficient; when $\vartheta^* \geq 0$, the chosen portfolio SSD-dominates the benchmark.

A key observation is that (5) assigns equal weight to every tail level: improving $\text{Tail}_{1/S}$ (worst single scenario) by δ contributes the same as improving $\text{Tail}_{S/S}$ (the mean) by δ . Fábíán et al. (2011b) address this by introducing a *scaled model* that weights each tail constraint by i/S , equivalently comparing CVaR values at confidence level i/S :

$$\max \vartheta \quad \text{subject to} \quad \frac{i}{S} \vartheta \leq \text{Tail}_{\frac{i}{S}}(R^x) - \hat{\tau}_i, \quad i = 1, \dots, S, \quad x \in X. \quad (6)$$

Because $\text{Tail}_{i/S}(R^x) = -\frac{i}{S} \text{CVaR}_{i/S}(R^x)$, constraint (6) is equivalent to $\text{CVaR}_{i/S}(R^x) \leq \text{CVaR}_{i/S}(R_I) - \vartheta$, so the scaled model maximises the uniform CVaR improvement across all confidence levels. Empirically, Roman et al. (2013) find that the scaled model tends to produce portfolios whose return distributions are shifted to the right of those from the unscaled model, yielding higher returns in most scenarios at the cost of slightly larger losses in the very worst ones. Both models are always feasible and always produce SSD efficient portfolios, regardless of the benchmark chosen.

2. Multi-Benchmark enhanced indexation based on stochastic dominance

Here we first review the multi-objective framework proposed in Roman et al. (2013) for identifying portfolios that second-order stochastically dominate, or approximately dominate, a given benchmark index. We then introduce a general approach for generating a new subset

of Pareto-optimal solutions within this multi-objective setting. In particular, we extend the minimax selection criterion previously discussed, while preserving the linear programming structure of the original model and providing more flexible mechanisms for selecting Pareto-optimal solutions from the resulting multi-objective formulation.

Among the infinitely many Pareto-optimal points belonging to the efficient T -dimensional hypersurface generated by Problem (7), the aforementioned framework selects a particular solution by solving the following minimax problem.

$$\begin{aligned} \max_x \quad & \min_{1 \leq j \leq S} \left(\text{Tail}_{\frac{j}{S}}(R(x)) - \text{Tail}_{\frac{j}{S}}(R_I) \right) \\ \text{s.t.} \quad & x \in C \end{aligned} \quad (7)$$

We note that the above problem is equivalent to

$$\begin{aligned} \min_x \quad & \left(\text{CVaR}_{\frac{j}{S}}(R(x)) - \text{CVaR}_{\frac{j}{S}}(R_I) \right) \\ \text{s.t.} \quad & x \in C \end{aligned} \quad (8)$$

This model can be expressed as a linear programming (LP) problem by using the CVaR representation proposed by (Rockafellar and Uryasev, 2000; Rockfeller and Uryasev, 2002). However, due to the exponential number of variables and constraints, namely $2^T - 1$, Problem (9) is typically solved by implementing cutting-plane techniques, as suggested by Fábíán et al. (2011a).

We consider an investment environment consisting of n available assets whose returns are represented by the random variables $R_1, \dots, R_n$. A portfolio is defined by the vector $x = (x_1, \dots, x_n) \in \mathbb{R}_+^n$, where x_k denotes the proportion of total capital allocated to the k -th asset. Consequently, the portfolio return can be expressed as $R(x) = \sum_{k=1}^n x_k R_k$.

Within this setting, Roman et al. (2013) introduce a multi-objective optimization model whose Pareto-efficient solutions achieve second-order stochastic dominance (SSD), or approximate SSD, with respect to a benchmark index. Related developments and applications of this framework can be found in (Roman et al., 2006; Fábíán et al., 2011a,b; Valle et al., 2017). The corresponding multi-objective formulation is given by:

$$\begin{aligned} \max_x \quad & \left[(\text{Tail}_{\frac{1}{S}}(R(x)) - \text{Tail}_{\frac{1}{S}}(R_I)), \dots, (\text{Tail}_{\frac{S}{S}}(R(x)) - \text{Tail}_{\frac{S}{S}}(R_I)) \right] \\ \text{s.t.} \quad & x \in C \end{aligned} \quad (9)$$

where C is a polyhedron. Alternatively, they find the SSD efficient portfolios by solving a slightly different multi-objective optimization problem based on CVaR .

$$\begin{aligned} \min_x \quad & \left[(\text{CVaR}_{\frac{1}{S}}(R(x)) - \text{CVaR}_{\frac{1}{S}}(R_I)), \dots, (\text{CVaR}_{\frac{S}{S}}(R(x)) - \text{CVaR}_{\frac{S}{S}}(R_I)) \right] \\ \text{s.t.} \quad & x \in C \end{aligned} \quad (10)$$

In particular, the formulation can be viewed as selecting a specific Pareto-efficient solution from the vector optimization problem in (7), where each component represents the difference between the tail expectation of the portfolio return and that of the benchmark distribution. In the following, we focus on Problems (9)–(10) and discuss how the associated minimax scalarization can be interpreted within the multi-objective framework described above.

2.1. Theoretical Results

REVISAR A PARTIR DAQUI

Inspired by the multi-objective approach of Roman et al. (2013), we can extend formulation (9) to the multi-benchmark setting. In this context, the proposed model seeks to construct portfolios that dominate multiple benchmark indices under the SSD criterion simultaneously. The approach relies on a minimax-type scalarization that evaluates the relative SSD performance of the candidate portfolio with respect to each benchmark. In particular, the method searches for portfolios that maximize the worst-case improvement in SSD dominance across the set of benchmark indices.

In a discrete setting with S equiprobable scenarios, SSD efficiency can be achieved by formulating a multi-objective problem in which the S objective functions—representing the unconditional expectations of the lowest $\frac{s}{S}$ outcomes of the portfolio return—are maximized.

We restrict our attention to the solution obtained through a minimax scalarization, which minimizes the worst deviation between the portfolio tail expectations and those of a set of benchmark distributions. In particular, the approach evaluates the portfolio performance relative to the worst benchmark tail among K different reference indices. Then let:

$$\max_{x \in \Delta} \min_{s \in \{1..S\}, k \in \{1..K\}} \left[Tail_{\frac{s}{S}}(R(x)) - Tail_{\frac{s}{S^k}}(R(I^k)) \right]$$

We can isolate the inner minimization over the multiple benchmarks:

$$\begin{aligned} &= \max_{x \in \Delta} \min_s \min_k \left[Tail_{\frac{s}{S}}(Rx) - Tail_{\frac{s}{S^k}}(R(I^k)) \right] \\ &= \max_{x \in \Delta} \min_t \left[Tail_{\frac{s}{S}}(Rx) + \min_k (-Tail_{\frac{s}{S^k}}(R(I^k))) \right] \end{aligned}$$

Since $\min(-f(x)) = -\max(f(x))$, we can rewrite the equation as:

$$= \max_{x \in \Delta} \min_t \left[Tail_{\frac{s}{S}}(Rx) - \max_k (Tail_{\frac{s}{S^k}}(R(I^k))) \right]$$

where $\Delta = \left\{ x \in \mathbb{R}^N \mid \sum_{n=1}^N x_n = 1, x_n \geq 0 \right\}$ is the set of feasible portfolios.

Hence, requiring a portfolio to SSD-dominate multiple benchmarks simultaneously is strictly equivalent to dominating a single, synthetic reference distribution. In this case, the target outcomes of this “synthetic benchmark” are defined by $\max (Tail_{\frac{s}{S^k}}(R(I^k)))$, meaning that, at each confidence level $\frac{s}{S}$, the model dynamically targets the most stringent tail requirement among all K available benchmarks.

The resulting single-objective optimization problem aims to maximize the worst “partial achievement” over the feasible set $x \in X$. Let $\hat{\tau}_s^k = Tail_{\frac{s}{S^k}}(I^k)$ denote the tail value

of the benchmark distribution I^k at level $\frac{s}{S^k}$, for $s = 1, \dots, S^k$ and $k = 1, \dots, K$. We define the auxiliary variable ϑ as the minimum difference between the tail expectation of the portfolio return and the largest corresponding benchmark tail value, that is, $\vartheta = \min_{s=1, \dots, S} \{ \text{Tail}_{\frac{s}{S}}(Rx) - \max_{k=1, \dots, K} \hat{\tau}_s^k \}$.

$$\max \quad \vartheta \tag{7}$$

$$\text{s.t.} \quad \vartheta \in \mathbb{R}, \quad x \in X, \tag{8}$$

$$\frac{i}{S} \vartheta \leq \frac{1}{S} \sum_{j \in \mathcal{J}_i} R(x) \quad \text{for each } \mathcal{J}_i \subset \{1, \dots, S\}, \quad |\mathcal{J}_i| = i, \tag{9}$$

$$\text{where } i = 1, \dots, S. \tag{8}$$

By imposing linear constraints that ensure this variable is greater than or equal to each $\hat{\tau}_s^k$, for $k = 1, \dots, K$, the nonlinear operator can be replaced by an equivalent linear representation. This reformulation preserves the original minimax structure of the problem while allowing the resulting model to be expressed as a linear optimization program, which can then be efficiently solved using standard linear programming techniques. We can reformulate the problem above as

$$\max \quad \vartheta \tag{7}$$

$$\text{subject to} \quad \vartheta \leq \vartheta_s^k, \quad s = 1, \dots, S^k, \quad k = 1, \dots, K \tag{8}$$

$$\vartheta_s^k \leq \frac{1}{S^k} \sum_{j \in \mathcal{J}} \sum_{i=1}^N r_{ij} w_i - \max(\hat{\tau}_s^k), \quad \mathcal{J} \subseteq \{1, \dots, S^k\}, \quad |\mathcal{J}| = s, \tag{9}$$

$$s = 1, \dots, S^k, \quad k = 1, \dots, K$$

$$\sum_{i=1}^N w_i = 1 \tag{10}$$

$$\vartheta_s^k \in \mathbb{R}, \quad s = 1, \dots, S^k, \quad k = 1, \dots, K \tag{11}$$

$$z \in \mathbb{R} \tag{13}$$

Alternatively, as demonstrated by Roman et al. (2013), finding SSD-efficient portfolios can be achieved by solving an equivalent multi-objective optimization problem based on Conditional Value-at-Risk (CVaR) at S different confidence levels. Therefore, we can shift from maximizing returns in the worst-case scenarios to minimizing the worst-case risk deviations. To ensure that the portfolio SSD-dominates multiple benchmarks, we minimize the maximum deviation of the portfolio's CVaR from the target CVaRs of the K benchmarks. Let $\hat{\tau}_s^k$ denote the target CVaR of benchmark k at confidence level $\frac{s}{S}$.

The minimax formulation is given by:

$$\min_{x \in \Delta} \max_{t \in \{1..T\}, k \in \{1..K\}} \left[\text{CVaR}_{\frac{s}{S}}(R^T x) - \hat{\tau}_s^k \right]$$

We can isolate the inner maximization over the multiple benchmarks:

$$\begin{aligned}
&= \min_{x \in \Delta} \max_t \max_k \left[\text{CVaR}_{\frac{s}{S}}(R^T x) - \hat{\tau}_s^k \right] \\
&= \min_{x \in \Delta} \max_t \left[\text{CVaR}_{\frac{s}{S}}(R^T x) + \max_k (-\hat{\tau}_s^k) \right]
\end{aligned}$$

Since $\max(-f(x)) = -\min(f(x))$, we can appropriately rewrite the equation as:

$$= \min_{x \in \Delta} \max_t \left[\text{CVaR}_{\frac{s}{S}}(R^T x) - \min_k (\hat{\tau}_s^k) \right]$$

where $\Delta = \left\{ x \in \mathbb{R}^N \mid \sum_{n=1}^N x_n = 1, x_n \geq 0 \right\}$ is the set of feasible portfolios.

While dominating multiple benchmarks in the outcome space requires maximise the minimum tail difference with respect to the synthetic benchmark, dominating them in the risk space requires targeting the most stringent risk profile among them. Thus, the synthetic benchmark is dynamically constructed by selecting the minimum target risk among all K benchmarks at every confidence level $\frac{s}{S}$. By transforming the problem into a CVaR minimization framework, we can efficiently compute the optimal portfolio utilizing standard LP formulations.

This formulation allows the decision-maker to incorporate several reference portfolios or market indices within a unified optimization framework, thereby generating portfolios that achieve robust stochastic dominance properties relative to multiple benchmarks. Consequently, the resulting solutions represent efficient trade-offs in the multi-benchmark dominance space and provide a flexible tool for portfolio construction in environments where several benchmarks are relevant for performance evaluation.

Remark 3 (Minimax Scalarization Vs. Multi-benchmark). From Problem (10), one can directly observe the link between Model (9) in the paper by Roman et al. (2013), SSD (Definition 3) and Multi-SSD (Definition 4). Indeed, denoting by x^* the optimal solution of Problem (10), if the corresponding optimal value function

$$(\text{CVaR}_{\frac{s}{S}}(R(x^*)) - \text{CVaR}_{\frac{s}{S}}(R_I))^{(1)}$$

is nonpositive (positive), then the corresponding optimal portfolio SSD-dominates (ASSD-dominates) the benchmark index. Indeed, by considering several benchmark distributions simultaneously, the tail differences can be evaluated with respect to multiple reference indices, leading to a generalized formulation in which the portfolio seeks to dominate, or approximately dominate, a set of benchmarks under the SSD criterion.

3. Cutting-plane based SSD formulation

Based on a reformulation of the CVaR minimisation problem by Künzi-Bay and Mayer (2006); Fábián et al. (2011a) proposed a novel cutting-plane formulation of the SSD problem,

one whose objective is to maximise the minimum value in the SSD relationship between the portfolio and a given benchmark (reference distribution). Roman et al. (2013) then employed the formulation for enhanced indexation. Let:

N	number of assets into which we may invest.
S	number of scenarios
$R = (R_1, \dots, R_S)^T$	discrete random variables denoting the returns of assets in $1, \dots, N$
I	discrete random variable denoting the returns of the benchmark
r_{is}	return of asset i under scenarios $s = 1, \dots, S$
r_s^I	benchmark return under scenario s
$Tail_\alpha^L(R)$	unconditional expectation of the smallest $\alpha \times 100\%$ outcomes of R

In the case of equiprobable scenarios, Fábíán et al. (2011a) defined, for a portfolio P , $Tail_{\frac{s}{S}}^L(P)$ as the left tail of the cumulative return distribution of P weighted by the constant $(1/S)$ factor:

$$Tail_{\frac{s}{S}}^L(P) = \frac{1}{S} \sum_{j=1}^s r_{(j)}^P$$

where $r_{(j)}^P$ denotes the j -th smallest portfolio return.

Let $\hat{\tau} = (\hat{\tau}_1, \dots, \hat{\tau}_S) = (Tail_{\frac{1}{S}}^L(I), \dots, Tail_{\frac{S}{S}}^L(I))$. Let w_i be the proportion to be invested in asset $i = 1, \dots, N$.

Inspired by the CVaR cutting-plane reformulation by Küenzi-Bay and Mayer (2006); Fábíán et al. (2011a) proposed the following redefinition of $Tail_{\frac{s}{S}}^L(P)$:

$$Tail_{\frac{s}{S}}^L(P) = \frac{1}{S} \min \sum_{j \in \mathcal{J}} \sum_{i=1}^N r_{ij} w_i \quad \text{such that } \mathcal{J} \subset \{1, \dots, S\} \quad |\mathcal{J}| = s.$$

Let $\mathcal{V}$ be an auxiliary variable denoting the minimum value in the SSD relationship between portfolio and benchmark.

The cutting-plane SSD formulation is given by:

$$\max \quad \mathcal{V} \tag{1}$$

$$\mathcal{V} \leq \frac{1}{S} \sum_{j \in \mathcal{J}} \sum_{i=1}^N r_{ij} w_i - \hat{\tau}_s \quad \forall \mathcal{J} \subset \{1, \dots, S\}, \quad |\mathcal{J}| = s, s = \{1, \dots, S\} \tag{2}$$

subject to

$$\sum_{i=1}^N w_i = 1 \tag{3}$$

$$\mathcal{V} \in \mathbb{R} \tag{4}$$

By solving (1) subject to (2)-(4), we find the portfolio that maximises the minimum value in the SSD relation.

If no dominating portfolio exists, the objective function value is negative and represents the portfolio that comes the closest to dominating the benchmark.

The separation of Constraints (2) is done via inspection:

1. Start with at least one Constraint of type (2) to prevent the formulation from being unbounded.
2. Solve the formulation to obtain portfolio $w^* = (w_1^*, \dots, w_N^*)$ with returns $r_s^P = \sum_{i=1}^N r_{is} w_i^*$.
3. Verify if $\mathcal{V} > \sum_{j=1}^s r_{(j)}^P$ for any $s = \{1, \dots, S\}$. If not, stop. Otherwise, add the corresponding violated constraint(s) to the model and go back to Step (2).

Given a benchmark, there may be multiple distinct dominating portfolios. The SSD formulation can be tailored to focus on certain aspects of the portfolio distribution. With Constraints (2), more importance is given to $Tail_{\frac{L}{S}}(P)$ when s is small. Namely, $Tail_{\frac{L}{S}}(P)$ for s approaching S is given the same relative importance by Constraints (2) as for s close to 1. But since the left tails are cumulative, for large values of s , the most positive portfolio returns are "diluted" among smaller returns. An unintended consequence of this is that solving the formulation tends to yield portfolios that have a smaller left tail when compared to the benchmark, but also a smaller right tail.

As an alternative Fábíán et al. (2011b) proposed scaling the tails by replacing Constraints (2) with:

$$\frac{s}{S} \mathcal{V} \leq \frac{1}{S} \sum_{j \in \mathcal{J}_s} \sum_{i=1}^N r_{ij} w_i - \hat{\tau}_s \quad \forall \mathcal{J}_s \subset \{1, \dots, S\}, |\mathcal{J}_s| = s, s = \{1, \dots, S\} \quad (5)$$

With Constraints (5), empirically optimal portfolio distributions tend to have longer right tails at the cost of having also slightly longer left tails.

An extended version of the cutting-plane formulation with other real-world constraints was proposed in Valle et al. (2017). Due to the use of binary variables the standard separation algorithm was reimplemented as a Branch-and-cut.

4. MultiSSD OWA Redefinition + Cutting Planes

5. Later: Link with Valle,2017

6. Empirical part

Dataset: SP500

Models to compare: ROMANcvar; TONIcvar; ROMANTail; TONITail; MinVariance; SP500 index; $I_1, I_2, \dots, I_m$.

References

- Carlo Acerbi and Dirk Tasche. On the coherence of expected shortfall. *Journal of banking & finance*, 26(7):1487–1503, 2002.
- Vijay S. Bawa. Stochastic dominance: A research bibliography. *Management Science*, 28(6): 698–712, 1982. doi: 10.1287/mnsc.28.6.698.
- Daniel Bernoulli. Exposition of a new theory on the measurement of risk. *Econometrica*, 22: 23–36, 1954. doi: 10.2307/1909829. Translation of the original Latin paper published in St. Petersburg in 1738.
- David Blackwell. Equivalent comparisons of experiments. *Annals of Mathematical Statistics*, 24(2):265–272, 1953. doi: 10.1214/aoms/1177729032.
- C. Fábián, G. Mitra, and D. Roman. Processing second-order stochastic dominance models using cutting-plane representations. *Mathematical Programming*, 130(1):33–57, 2011a. doi: 10.1007/s10107-009-0326-1.
- C. Fábián, G. Mitra, D. Roman, and V. Zverovich. An enhanced model for portfolio choice with SSD criteria: a constructive approach. *Quantitative Finance*, 11(10):1525–1534, 2011b. doi: 10.1080/14697680903493607.
- Josef Hadar and William R. Russell. Rules for ordering uncertain prospects. *American Economic Review*, 59(1):25–34, 1969.
- Giora Hanoch and Haim Levy. The efficiency analysis of choices involving risk. *Review of Economic Studies*, 36(3):335–346, 1969. doi: 10.2307/2296431.
- Yoram Kroll and Haim Levy. Stochastic dominance: A review and some new evidence. *Research in Finance*, 2(1):163–227, 1980.
- Alexandra Küenzi-Bay and János Mayer. Computational aspects of minimizing conditional value-at-risk. *Computational Management Science*, 3(1):3–27, 2006. doi: 10.1007/s10287-005-0042-0.
- Haim Levy. Stochastic dominance and expected utility: survey and analysis. *Management Science*, 38(4):555–593, 1992. doi: 10.1287/mnsc.38.4.555.
- Haim Levy. *Stochastic dominance: Investment decision making under uncertainty*. Springer, 2006.
- R. T. Rockafellar and S. Uryasev. Optimization of conditional Value-at-Risk. *Journal of Risk*, 2:21–41, 2000. doi: 10.21314/JOR.2000.038.
- T Rockfeller and Stanislav Uryasev. Conditional value-at-risk for general loss distribution. *Journal of BANKING and FINANCE*, 26(7):1443–71, 2002.

- D. Roman, K. Darby-Dowman, and G. Mitra. Portfolio construction based on stochastic dominance and target return distributions. *Mathematical Programming*, 108(2):541–569, 2006. doi: 10.1007/s10107-006-0722-8.
- D. Roman, G. Mitra, and V. Zverovich. Enhanced indexation based on second-order stochastic dominance. *European Journal of Operational Research*, 228(1):273–281, 2013. doi: 10.1016/j.ejor.2013.01.035.
- Michael Rothschild and Joseph E. Stiglitz. Increasing risk: I. a definition. *Journal of Economic Theory*, 2(3):225–243, 1970. doi: 10.1016/0022-0531(70)90038-4.
- C. A. Valle, D. Roman, and G. Mitra. Novel Approaches for Portfolio Construction Using Second Order Stochastic Dominance. *Computational Management Science*, 14(2):257–280, 2017. doi: 10.1007/s10287-017-0274-9.
- John von Neumann and Oskar Morgenstern. *Theory of Games and Economic Behavior*. Princeton University Press, Princeton, NJ, 2nd edition, 1947.
- G. A. Whitmore. Third-degree stochastic dominance. *American Economic Review*, 60(3): 457–459, 1970.
- George A Whitmore and M Chapman Findlay. Stochastic dominance: an approach to decision-making under risk. (*No Title*), 1978.